

MOHAMMED ECH-CHAIBI

Automation & Mechatronics Engineer

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PROFESSIONAL SUMMARY

I build systems that work — from autonomous robotic harvesters to aerospace helium modelling for a next-generation airship. My background spans PLC programming, machine vision, industrial robotics, and process engineering across three countries and two engineering degrees. I am looking for large-scale, complex automation and controls engineering projects where the work is hard and the impact is real.

WORK HISTORY

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|--|-------------------------------|
| Automation engineer, University of Limerick | 12/2025 – Present
Limerick |
| <ul style="list-style-type: none">• Commissioned and integrated an industrial robotic harvesting cell combining a Mitsubishi RV-6S robot, CompactLogix/GuardLogix PLC, FactoryTalk HMI, TCP/IP communication, pneumatics, industrial I/O, and ISO 10218 safety functions.• Programmed the robot in MELFA BASIC IV, implementing Ethernet socket communication, PLC-driven command execution, synchronized robot-PLC handshaking, and autonomous/manual operating modes.• Co-designed and integrated a custom 3D-printed harvesting end-effector with pneumatic actuation, cutting shears, camera mounting, and a Raspberry Pi/OpenCV vision system for lettuce detection and harvest triggering. | |
| Automation engineer, University of Limerick | 10/2025 – 12/2025
Limerick |
| <ul style="list-style-type: none">• Programmed an Allen-Bradley Studio 5000 PLC to execute a 6-step cube inspection sequence — orientation detection, motor-driven rotation, face imaging, and return-to-home — running reliably with full HMI visibility and alarm handling.• Designed HMI screens covering operation modes, status monitoring, alarms, and manual override, following GAMP 5 good practice principles.• Implemented IEC 61508-aligned safety functions including E-stop, pause/resume logic, interlocks, and structured alarm management. | |
| Automation engineer, University of Limerick | 02/2026 – 03/2026
Limerick |
| <ul style="list-style-type: none">• Designed and commissioned a conveyor-based bottle sorting system integrating through-beam and proximity sensors, a pneumatic cylinder, and solenoid valve — achieving consistent and reliable rejection accuracy throughout all test cycles.• Developed LabVIEW/myRIO control logic and built the complete system including conveyor wiring, sensor integration, HMI bottle counting, and emergency stop functionality. | |
| Intern - manufacturing engineer, Flying Whales | 04/2025 – 09/2025
Bordeaux |
| <ul style="list-style-type: none">• Automated helium mass calculations across 50 assembly stages using Excel and Python, reducing verification time from hours to minutes and making the model instantly adaptable to future design changes.• Developed original algorithms to compute helium fill height across 14 independent cells and introduced a systematic imbalance-detection method, replacing manual mass comparison with a threshold-based check against a 2-tonne buoyancy target per cell. | |

- Analysed factory temperature and pressure variations on helium volume behaviour and initiated a risk assessment, identifying key failure scenarios and producing engineering cautions to strengthen the robustness of the filling sequence.

Intern -Automation and Process Improvement, SITI TEA

05/2024 – 09/2024

Marrakech

- Performed PID tuning on an 8-balance dosing system handling multiple tea product variants, reducing material waste from up to 7% down to 2–5% on the most complex heterogeneous materials.
- Identified material loss at the dosing table through Lean analysis and proposed a vibration and corner recovery system — recommendation formally accepted by management and flagged for future production projects.
- Developed standardised work instructions and proposed layout optimisations to eliminate non-value-added activities and simplify operator workflow.

EDUCATION

Master of engineering: Mechatronics, UNIVERSITY OF LIMERICK

09/2025 – 08/2026

LIMERICK

- *Robotics, automation, control systems, embedded systems, machine vision — with practical work in industrial robotics, PLCs, sensors, and system integration.*

Engineering Degree: Industrial Engineering, IMT NORD EUROPE

09/2023 – 08/2025

Douai

- *Industry 4.0 systems thinking, process optimisation, automation, production flow analysis and project management.*

CPGE: Engineering, Ibn Timiya

09/2021 – 09/2023

Marrakech

- *Two-year intensive preparatory programme — advanced mathematics, physics, and engineering science, equivalent to first two years of an engineering degree.*

SKILLS

Automation & PLC

Allen-Bradley Studio 5000, Siemens TIA Portal, State machines & structured sequence design, Emergency stop logic, safety interlocks, Sensor/actuator control&wiring basics.

Software&Tools

Python, C/C++,sensor integration, MATLAB, Git, Serial TCP communication.

Robotics

Mitsubishi RT Toolbox/Melfa Basic, Robot calibration, Robot-PLC communication, 3D Modeling.

Industrial engineering

Lean principles, process analysis, KPI dashboarding, workflow optimisation.

LANGUAGES

French

Fluent

English

Fluent

Spanish

Elementary

HOBBIES AND INTERESTS

- Chess
- Cooking
- Tennis

CERTIFICATIONS

- CSWA in 3D Modelling
- Lean Six Sigma Green Belt
- MOOC in Project Management.